Nama = Tsabita Rosyidah Putri

NPM = 22083010012

Memasukkan data

In [34]:

```
import pandas as pd
df = pd.read_csv('Sample-Superstore.xls.csv')
df
```

Out[34]:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	...	Postal Code
0	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	42420
1	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	42420
2	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	...	90036
3	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	33311
4	5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	33311
...	...	...	...	...	...	...	...	...	...	...	...	...
9989	9990	CA-2014-110422	1/21/2014	1/23/2014	Second Class	TB-21400	Tom Boeckenhauer	Consumer	United States	Miami	...	33180
9990	9991	CA-2017-121258	2/26/2017	3/3/2017	Standard Class	DB-13060	Dave Brooks	Consumer	United States	Costa Mesa	...	92627
9991	9992	CA-2017-121258	2/26/2017	3/3/2017	Standard Class	DB-13060	Dave Brooks	Consumer	United States	Costa Mesa	...	92627
9992	9993	CA-2017-121258	2/26/2017	3/3/2017	Standard Class	DB-13060	Dave Brooks	Consumer	United States	Costa Mesa	...	92627
9993	9994	CA-2017-119914	5/4/2017	5/9/2017	Second Class	CC-12220	Chris Cortes	Consumer	United States	Westminster	...	92683

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	...	Postal Code

9994 rows × 21 columns

In [35]:

```
df2 = df[['State','Profit','Sales']]  #Filter data
df2
```

Out[35]:

	State	Profit	Sales
0	Kentucky	41.9136	261.9600
1	Kentucky	219.5820	731.9400
2	California	6.8714	14.6200
3	Florida	-383.0310	957.5775
4	Florida	2.5164	22.3680
...	...	...	...
9989	Florida	4.1028	25.2480
9990	California	15.6332	91.9600
9991	California	19.3932	258.5760
9992	California	13.3200	29.6000
9993	California	72.9480	243.1600

9994 rows × 3 columns

Berapa angka rata - rata penjualan dan laba di california?

In [41]:

```
df3 = df2[df2['State'] == 'California']
```

In [42]:

```
df4 = df3.groupby('State')[['Profit', 'Sales']].mean().reset_index()
df4
```

Out[42]:

State	Profit	Sales
California	38.171608	228.729451

Negara bagian mana yang memiliki total penjualan tertinggi dan terendah?

In [39]:

```
df5 = df.groupby('State')['Sales'].sum().sort_values(ascending = False).reset_index()
df5
```

Out[39]:

	State	Sales
0	California	457687.6315
1	New York	310876.2710
2	Texas	170188.0458

	State	Sales
3	Washington	138641.2700
4	Pennsylvania	116511.9140
5	Florida	89473.7080
6	Illinois	80166.1010
7	Ohio	78258.1360
8	Michigan	76269.6140
9	Virginia	70636.7200
10	North Carolina	55603.1640
11	Indiana	53555.3600
12	Georgia	49095.8400
13	Kentucky	36591.7500
14	New Jersey	35764.3120
15	Arizona	35282.0010
16	Wisconsin	32114.6100
17	Colorado	32108.1180
18	Tennessee	30661.8730
19	Minnesota	29863.1500
20	Massachusetts	28634.4340
21	Delaware	27451.0690
22	Maryland	23705.5230
23	Rhode Island	22627.9560
24	Missouri	22205.1500
25	Oklahoma	19683.3900
26	Alabama	19510.6400
27	Oregon	17431.1500
28	Nevada	16729.1020
29	Connecticut	13384.3570
30	Arkansas	11678.1300
31	Utah	11220.0560
32	Mississippi	10771.3400
33	Louisiana	9217.0300
34	Vermont	8929.3700
35	South Carolina	8481.7100
36	Nebraska	7464.9300
37	New Hampshire	7292.5240
38	Montana	5589.3520
39	New Mexico	4783.5220
40	Iowa	4579.7600
41	Idaho	4382.4860
42	Kansas	2914.3100
43	District of Columbia	2865.0200
44	Wyoming	1603.1360
45	South Dakota	1315.5600
46	Maine	1270.5300
47	West Virginia	1209.8240

	State	Sales
48	North Dakota	919.9100

Total penjualan tertinggi

In [17]:

```
df6 = df5.groupby('State')['Sales'].sum().sort_values().reset_index()
(df6[-1:])
```

Out[17]:

	State	Sales
48	California	457687.6315

Total penjualan terendah

In [19]:

```
df6 = df5.groupby('State')['Sales'].sum().sort_values().reset_index()
(df6[:1])
```

Out[19]:

	State	Sales
0	North Dakota	919.91

Segmen konsumen apa yang memiliki variasi paling banyak dalam penjualan atau laba?

In [21]:

```
df_2 = df[['Segment','Profit','Sales']]  #Sort
df_2
```

Out[21]:

	Segment	Profit	Sales
0	Consumer	41.9136	261.9600
1	Consumer	219.5820	731.9400
2	Corporate	6.8714	14.6200
3	Consumer	-383.0310	957.5775
4	Consumer	2.5164	22.3680
...	...	...	...
9989	Consumer	4.1028	25.2480
9990	Consumer	15.6332	91.9600
9991	Consumer	19.3932	258.5760
9992	Consumer	13.3200	29.6000
9993	Consumer	72.9480	243.1600

9994 rows × 3 columns

In [29]:

```
v = df_2.groupby('Segment')[['Sales', 'Profit']].var().reset_index()
v1 = v.rename(columns={'Sales' : 'Varian Penjualan', 'Profit' : 'Varian Laba'})
```

In [31]:

```
spt = v1['Segment'][v1['Varian Penjualan'].idxmax()]
slt = v1['Segment'][v1['Varian Laba'].idxmax()]
```

In [32]:

```
print("Segmen konsumen dengan variasi penjualan tertinggi adalah ", spt)
print("Segmen konsumen dengan variasi laba tertinggi adalah ", slt)
```

```
Segmen konsumen dengan variasi penjualan tertinggi adalah  Home Office
Segmen konsumen dengan variasi laba tertinggi adalah  Consumer
```

Diantara 5 negara bagian teratas dalam penjualan, yaitu mode pengiriman dan kategori produk pilihan yang paling populer?

In [95]:

```
df5.head(5)
```

Out[95]:

	State	Sales
0	California	457687.6315
1	New York	310876.2710
2	Texas	170188.0458
3	Washington	138641.2700
4	Pennsylvania	116511.9140

In [97]:

```
df_ = df[['State','Ship Mode','Category']]
df_
```

Out[97]:

	State	Ship Mode	Category
0	Kentucky	Second Class	Furniture
1	Kentucky	Second Class	Furniture
2	California	Second Class	Office Supplies
3	Florida	Standard Class	Furniture
4	Florida	Standard Class	Office Supplies
...	...	...	...
9989	Florida	Second Class	Furniture
9990	California	Standard Class	Furniture
9991	California	Standard Class	Technology
9992	California	Standard Class	Office Supplies
9993	California	Second Class	Office Supplies

9994 rows × 3 columns

In [99]:

```
State = ['California', 'New York', 'Texas', 'Washington', 'Pennsylvania']

dp = df_[(df_['State'].isin(State))]
dp
```

Out[99]:

	State	Ship Mode	Category
2	**California**	**Second Class**	**Office Supplies**
5	**California**	**Standard Class**	**Furniture**
6	**California**	**Standard Class**	**Office Supplies**
7	**California**	**Standard Class**	**Technology**
8	**California**	**Standard Class**	**Office Supplies**
...	**...**	**...**	**...**
9986	**California**	**Standard Class**	**Technology**
9990	**California**	**Standard Class**	**Furniture**
9991	**California**	**Standard Class**	**Technology**
9992	**California**	**Standard Class**	**Office Supplies**
9993	**California**	**Second Class**	**Office Supplies**

5207 rows × 3 columns

In [100]:

```
dp.value_counts()
```

Out[100]:

```
State          Ship Mode       Category
California     Standard Class  Office Supplies    683
New York       Standard Class  Office Supplies    418
Texas          Standard Class  Office Supplies    387
California     Standard Class  Furniture          260
               Second Class    Office Supplies    246
Pennsylvania   Standard Class  Office Supplies    223
California     Standard Class  Technology         222
               First Class     Office Supplies    190
Washington     Standard Class  Office Supplies    183
New York       Standard Class  Furniture          143
               Second Class    Office Supplies    123
Texas          Standard Class  Furniture          120
New York       Standard Class  Technology         117
               First Class     Office Supplies    111
Texas          Second Class    Office Supplies    102
               Standard Class  Technology          95
               First Class     Office Supplies     84
California     Second Class    Furniture           83
               Same Day        Office Supplies     79
Pennsylvania   Standard Class  Technology          78
California     First Class     Furniture           72
Pennsylvania   Standard Class  Furniture           70
               First Class     Office Supplies     67
California     Second Class    Technology          66
Washington     Standard Class  Furniture           60
California     First Class     Technology          60
Washington     Standard Class  Technology          51
               Second Class    Office Supplies     50
New York       Second Class    Furniture           50
Texas          Second Class    Technology          46
Pennsylvania   Second Class    Office Supplies     43
Texas          Second Class    Furniture           43
Washington     First Class     Office Supplies     40
New York       First Class     Technology          40
               Second Class    Technology          36
               Same Day        Office Supplies     32
Washington     Second Class    Furniture           32
Texas          Same Day        Office Supplies     31
               First Class     Furniture           30
                               Technology          30
California     Same Day        Furniture           29
New York       First Class     Furniture           28
```

```
Pennsylvania  First Class     Furniture           26
              Second Class    Furniture           25
                              Technology          23
Washington    Second Class    Technology          23
              Same Day        Office Supplies     19
Pennsylvania  First Class     Technology          16
New York      Same Day        Furniture           15
                              Technology          15
Washington    First Class     Furniture           14
                              Technology          14
              Same Day        Technology          12
California    Same Day        Technology          11
Pennsylvania  Same Day        Office Supplies     10
Texas         Same Day        Furniture            9
                              Technology           8
Washington    Same Day        Furniture            8
Pennsylvania  Same Day        Furniture            4
                              Technology           2
dtype: int64
```

In [104]:

```
dp['Ship Mode'].value_counts()
```

Out[104]:

```
Standard Class    3110
Second Class       991
First Class        822
Same Day           284
Name: Ship Mode, dtype: int64
```

In [106]:

```
dp['Category'].value_counts()
```

Out[106]:

```
Office Supplies    3121
Furniture          1121
Technology          965
Name: Category, dtype: int64
```

In [ ]:

```

```